IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

ANALOGUE SIGNAL PROCESSING

Monday, 22 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E. Rodriguez-Villegas

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} : Drain Source Voltage.

V_{GS} : Gate Source Voltage.

V_{TH} : Threshold Voltage.

U_t : Thermal Voltage.

n : Weak Inversion Slope factor.

g_m: Transconductance.

r_o: Output channel resistance.

1. a) State two advantages and two disadvantages of using analogue circuits for signal processing.

[4]

b)

(i) Figure 1.1 shows a circuit which encodes a continuous-time signal to a different domain. Identify what domain the circuit is operating in and explain the circuit's operation showing time domain plots on how the amplitude is encoded; use equations where appropriate.

[6]

(ii) State two advantages of encoding signals in this domain.

[2]

c) Figure 1.2 shows a 3-stage amplifier-cascade. Derive the output referred noise equation for this system and using it, propose an effective way to reduce the overall noise of the system. *(You should state the gain expressions using g_m and r_o.)*

[8]

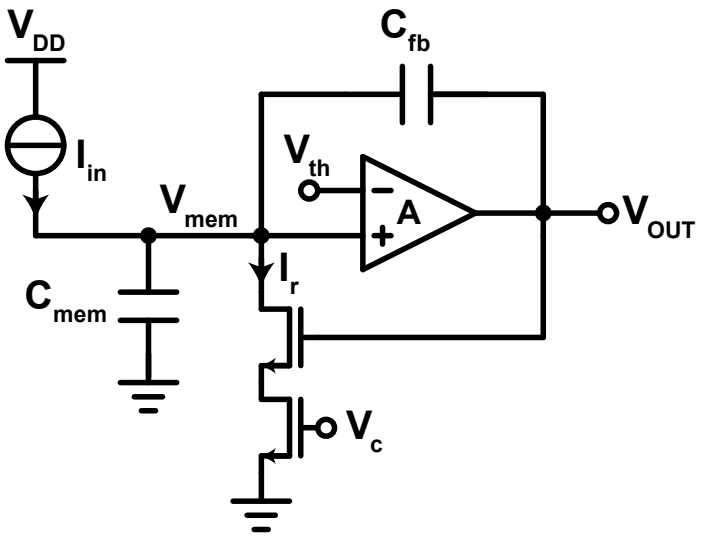


Figure 1.1

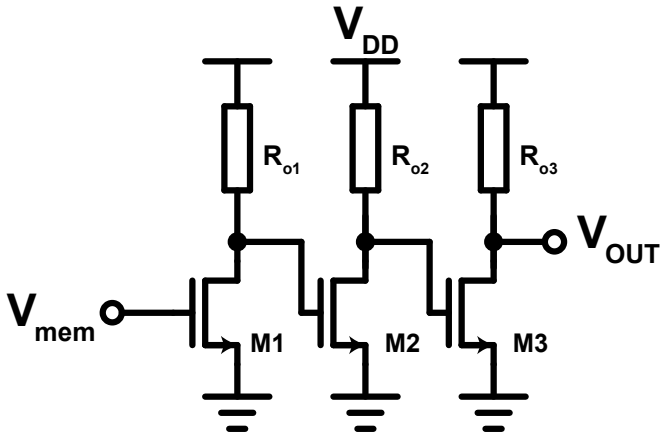


Figure 1.2

2. a) State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion.

[4]

b) Draw three devices which could be used as tranlinear elements justifying your choices.

[3]

c) Figure 2.1 shows a translinear circuit made up of NMOS transistors biased in weak inversion.

(i) Derive I_{out} as a function of X by applying the translinear principle. Show all necessary steps.

[4]

(ii) What is the function of the circuit shown in Figure 2.1 ?

[1]

d) The equation for a vector normaliser circuit is given as:

$$u = \frac{x}{\sqrt{x^2 + y^2}}$$

Synthesise a translinear circuit to provide this function minimsing the circuit where possible. You may only use Type A cells.
Hint: The function may be decomposed into the functions $u = \frac{x}{r}$ and $r = \sqrt{x^2 + y^2}$.

[8]

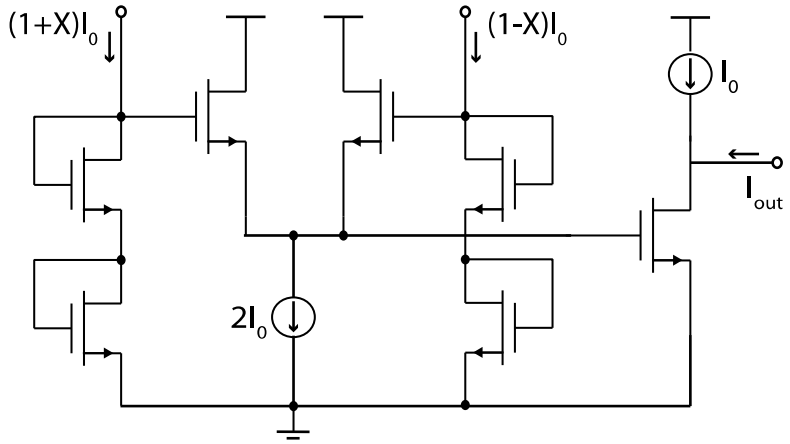


Figure 2.1

3. The transfer function of a second order topology has been decomposed into the following state-space equations:

$$\dot{X}_1 = -\frac{\omega_0}{Q} X_1 + \omega_0 X_2$$

$$\dot{X}_2 = -\omega_0 X_1 + \omega_0 U$$

$$Y_1 = X_1$$

$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs and X_1 and X_2 are the state-variables and U is the input.

a. Show that the output Y_1 can be used to implement a second order lowpass transfer function and the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

[4]

b. By using the mappings below, show how these state space equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

[6]

c. With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y_1 and Y_2 using weak inversion MOS transistors.

[8]

d. Given that the low pass biquad filter output at Y_1 has a cut-off frequency, f_0 = 4 kHz, select a suitable value for the tuning current I_ω given that the filter capacitor is C = 10 pF, n = 1.23 and U_t = 25 mV.

[2]

4. a) Figure 4.1 depicts a switched current memory cell.

i) State two limitations of the circuit shown in figure 4.1 and draw an improved schematic which compensates for these.

[4]

ii) Using your schematic from part 4.a.i, design a memory cell that is non-inverting.

[3]

iii) Figure 4.2 depicts a switched capacitor memory cell capable of error compensation. Explain its operation, stating the drain currents of M1 and M2 during each phase of the clock and derive the compensated output current.

[4]

iv) Figure 4.3 depicts a switched current cell. Derive its input/output characteristic and identify which function this block performs. You are required to state the transfer function in the z-domain.

[5]

b) This question concerns current conveyors. Design, using CCII+ current conveyors, a voltage mode and a current mode differentiator.

[4]

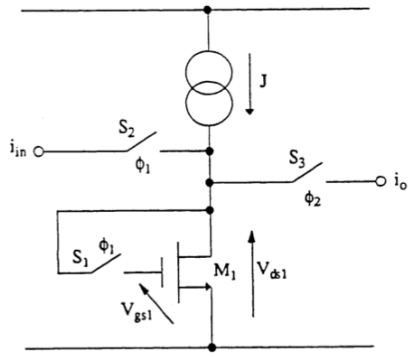


Figure 4.1

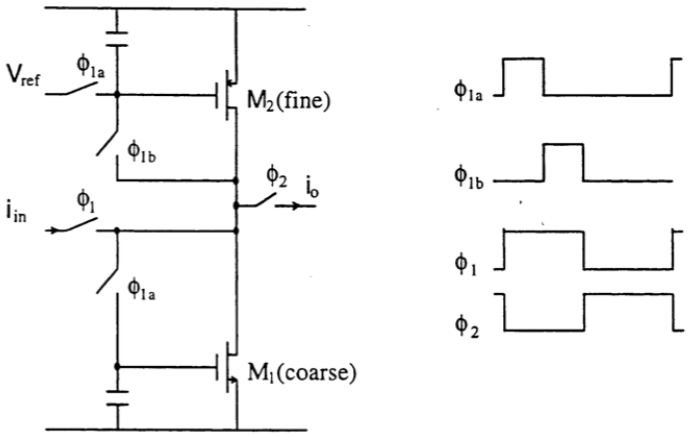


Figure 4.2

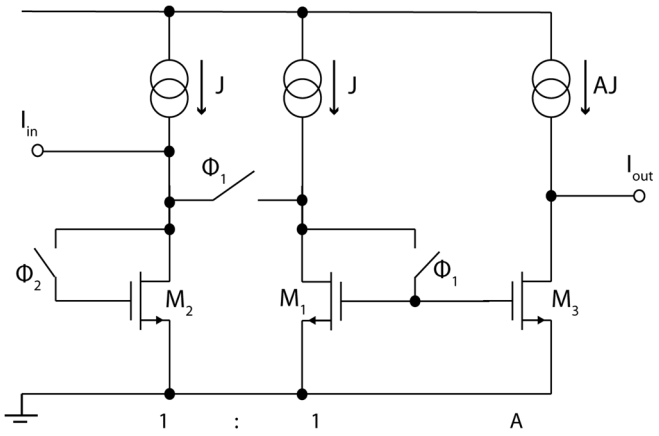


Figure 4.3

5. This question is on Switched-Capacitor Circuits.

a) (i) Fig. 5.1 shows a typical implementation of a switched-capacitor unit cell. Define its voltage range and speed limitations. Use equations as well as $V_{OUT}\ vs.time(t)$ and $R_{ON}\ vs.V_{OUT}$ plots to aid your explanation. *(Note: You may assume an initial condition of the node voltages to aid you explanation.)*

[3]

(ii) Draw an improved design of the switch and state how you may size the transistor(s) to address the previous limitations.

[2]

(iii) Explain why charge-injection is a non-ideal effect of using MOSFETs as switches for switched-capacitor unit cells. Draw and state the gain, offset and non-linearity issue due to charge injection when body effect is considered and briefly define three other non-idealities of MOSFET as switches for Switched Capacitor cells and write down the total voltage variation (ΔV) contributed from all four, including charge-injection. *(Note: You may assume all the channel charges are flowing towards C_H.) (Body effect is given by: $V_{TH} = V_{TH0} + \gamma\sqrt{2\phi_B + V_{SB}} + \gamma\sqrt{\phi_B}$)*

[5]

b) (i) Explain the operating principles of the circuits Fig.5.2(a) and (b) and state what improvement the circuit in 5.2(b) achieves compared to that in 5.2(a).

[5]

(ii) Derive and explain the functionality of the circuit shown in Fig 5.3.

[3]

(iii) Draw a revised circuit of Fig 5.3 which facilitates wideband operation.

[2]

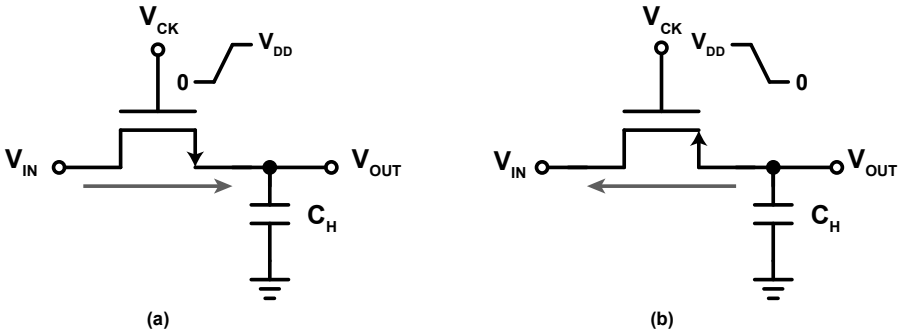


Figure 5.1

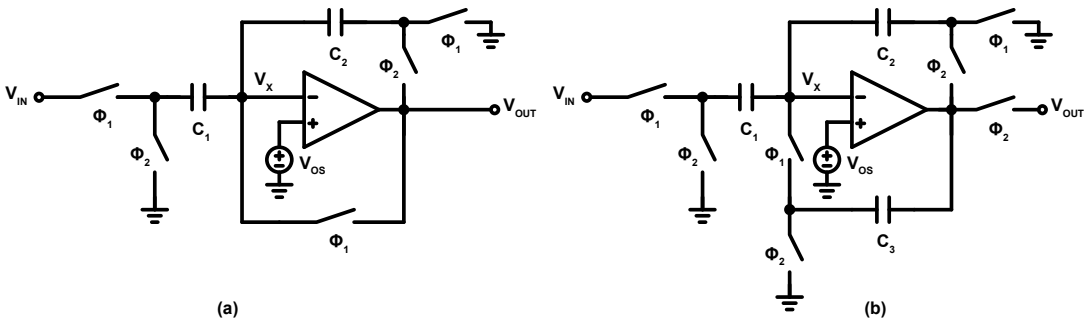


Figure 5.2

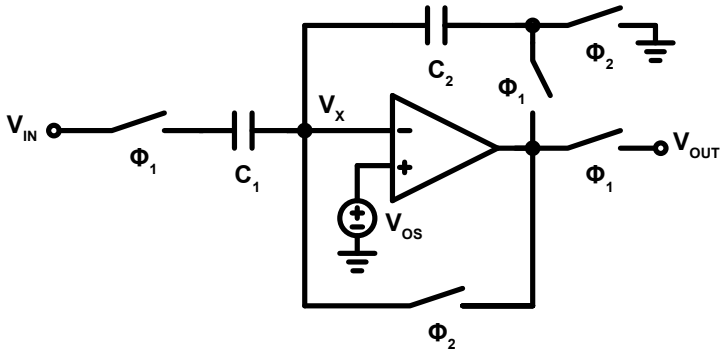


Figure 5.3

6.

(a) (i) Draw the block diagram of an auto-zeroing closed-loop amplifier with offset compensation using an auxiliary input and explain its operating principle.

[3]

(ii) Complete the circuit in Fig. 6.1 to apply the auxiliary-input offset compensation described in 6.a.i.

[3]

(b) (i) Fig. 6.2 shows a typical analogue signal processing chain that continuously amplifies an output signal from a differentially configured sensor pair to a single input ADC. You are responsible to design the missing circuits in the dotted box using the following criteria:

1. At least two-stages of amplification are required followed by an offset-free switched capacitor integrator.

2. The sensor's output has a frequency range of 1–10 Hz, and a signal magnitude of 10 mV. Each op-amp has a 1/f noise amplitude of 10 mV at 10 Hz, and a corner frequency of 1 kHz.

3. You should design the circuit so that the sensor's output is not buried by the input referred noise of your circuit.

4. You are only allowed to use the following symbols as listed in Fig 6.3 in addition to any voltage sources requred. You do not need to show the inside schematics of the op-amp itself.

[6]

(ii) Explain the method you used to achieve a good noise performance in your sensing system and also explain the operation of the offset-free switched capacitor integrator. You may use frequency spectrum plots to aid your description.

[8]

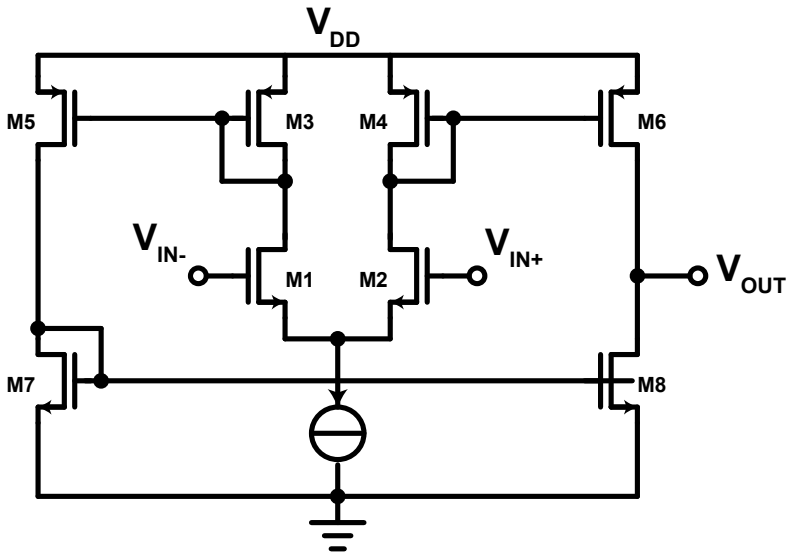


Figure 6.1

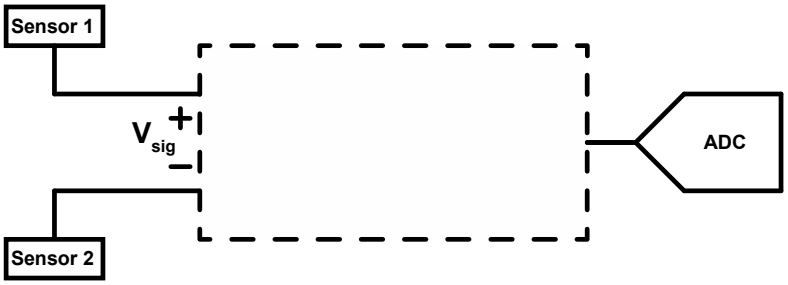


Figure 6.2

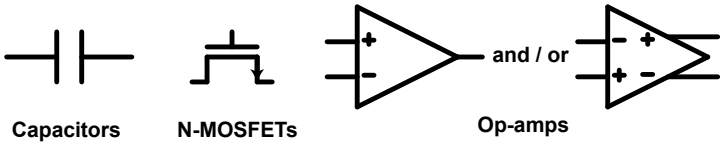


Figure 6.3